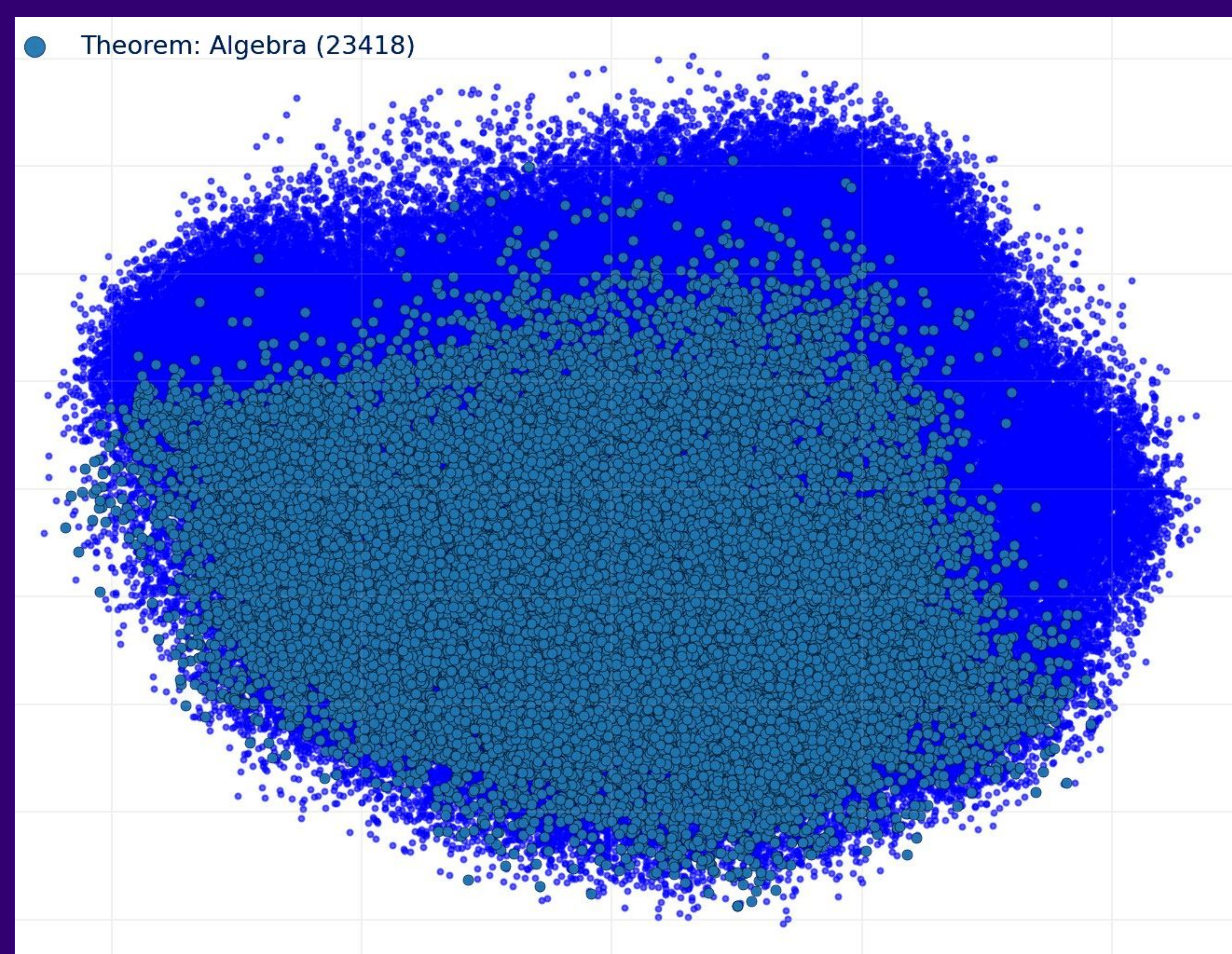
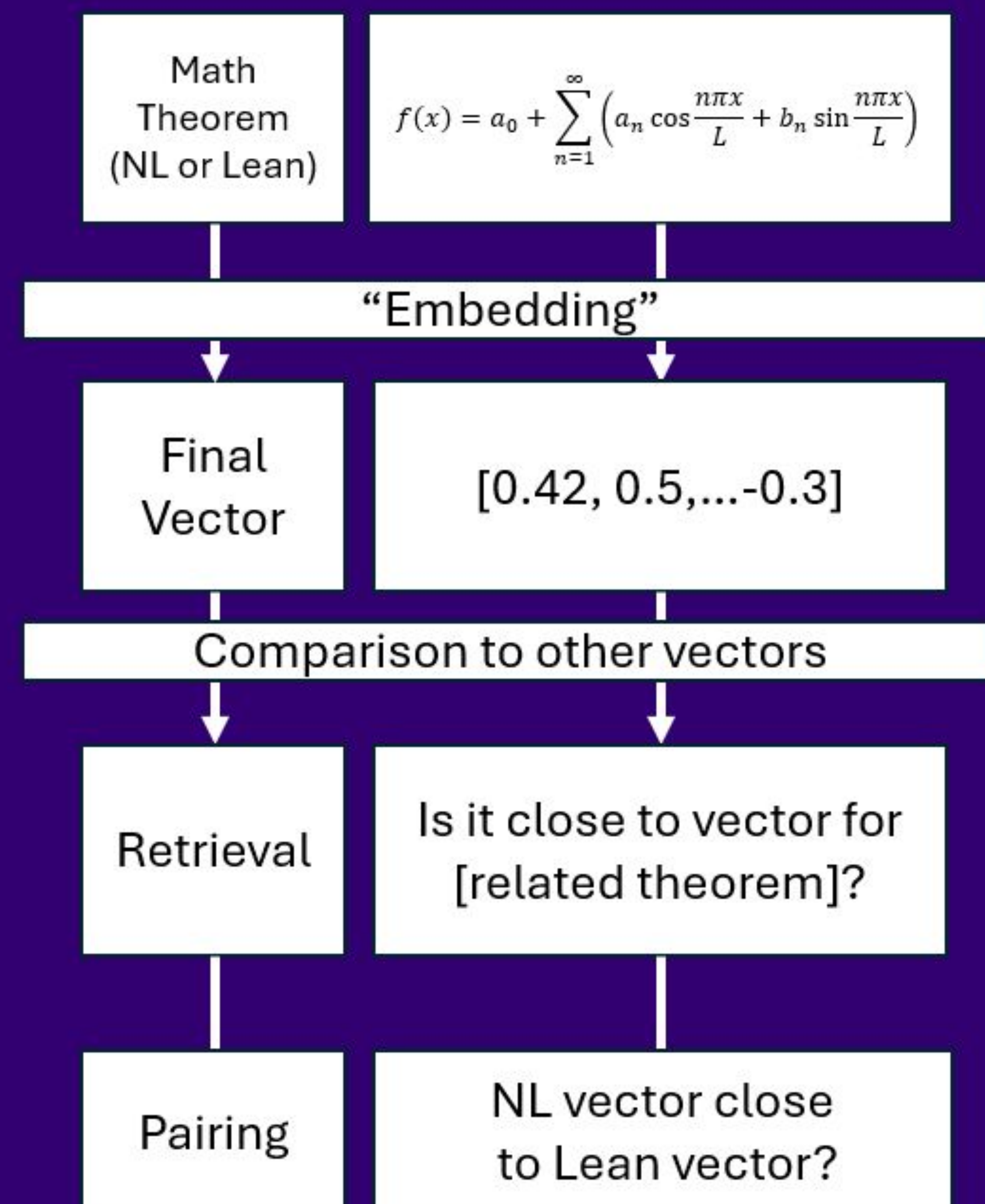


Math Embedding Pipeline



Math2Vec: Developing Embedding Models Capable of Capturing Mathematical Meaning

Samarth Rao, Saharsh Bhargava, Cecilia Ye, Jiahe Lu, Leo Carlin, Jared Darlington, Michael Zhou, Rachit Jaiswal, Henry Kvinge

What Is Text Embedding?

- An embedding converts a piece of text into a high-dimensional vector such that 'similar' text maps to vectors that are close together (in terms of cosine similarity).
- Once a body of text is realized as vectors, we can use basic tools from geometry (e.g., nearest neighbor algorithms) to search and analyze the text database

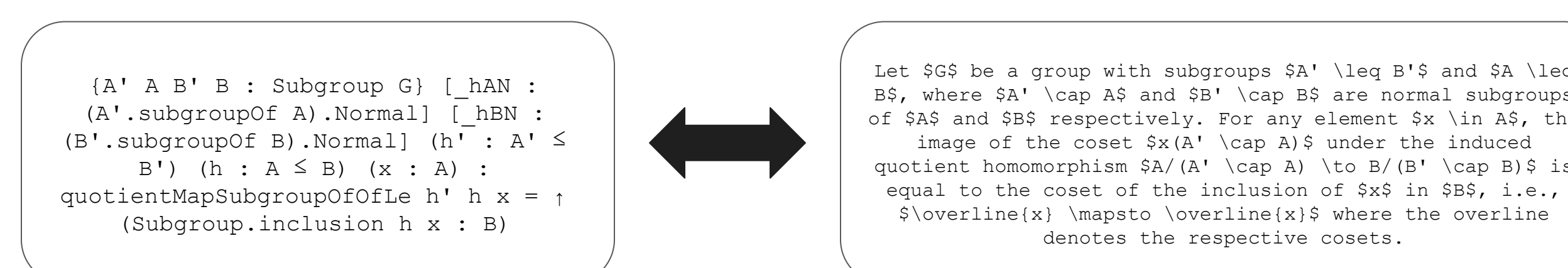
Why Are Embedding Models Useful in Mathematics?

Possible tasks include:

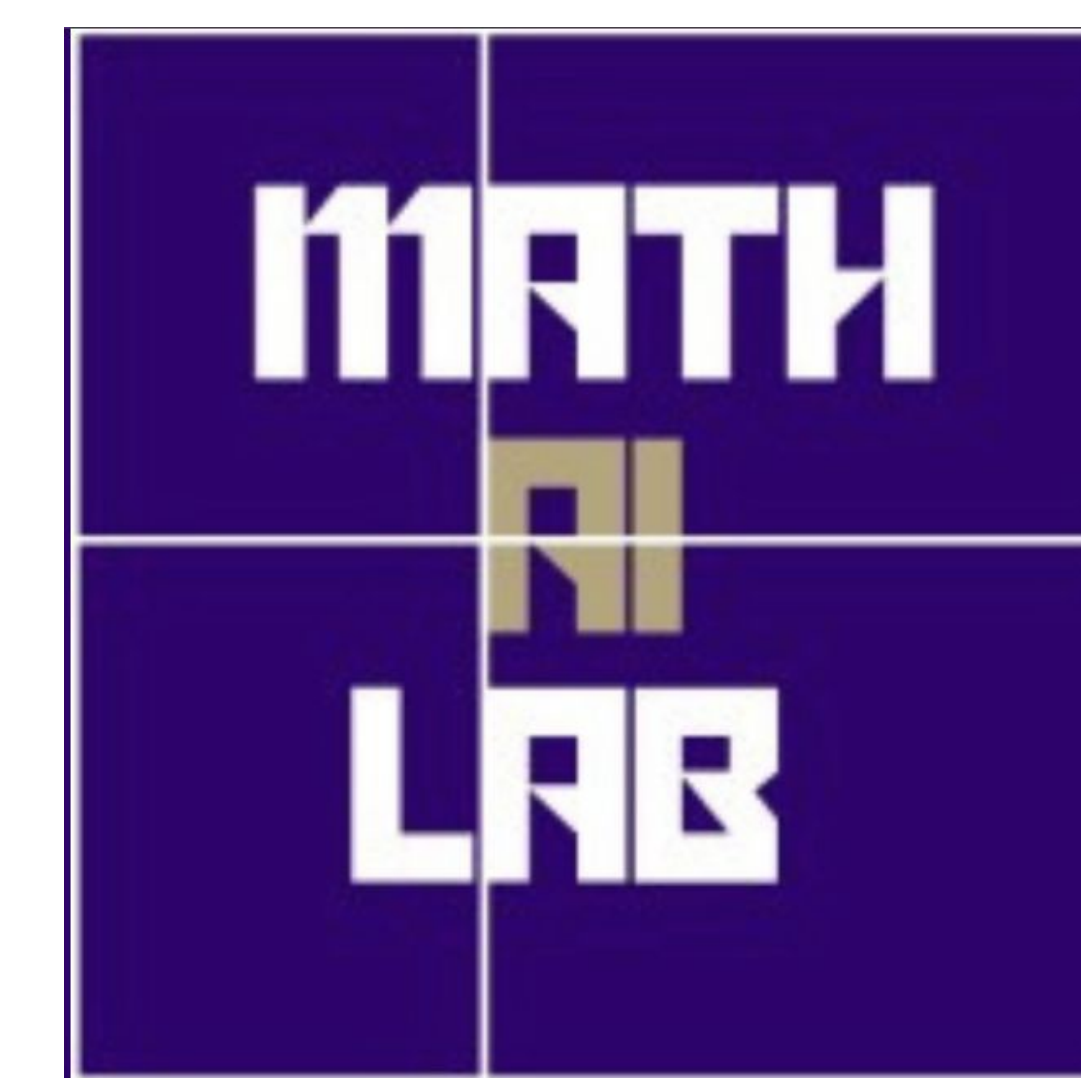
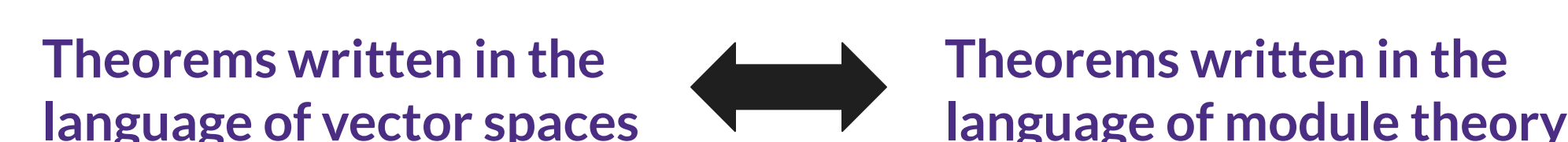
- **Find related results:** Given a newly proven theorem or proposed conjecture, search for related results
- **Natural language to Lean search:** Given a theorem in natural language, find the closest match in a Lean database
- **Analysis of mathematics:** Using embedding models, we can study large mathematical text corpora to understand the field

Datasets

- **Mathlib4 to natural text:** A dataset including Lean statements paired with a natural language translation
FrenzyMath/mathlib_informal v4.16.0



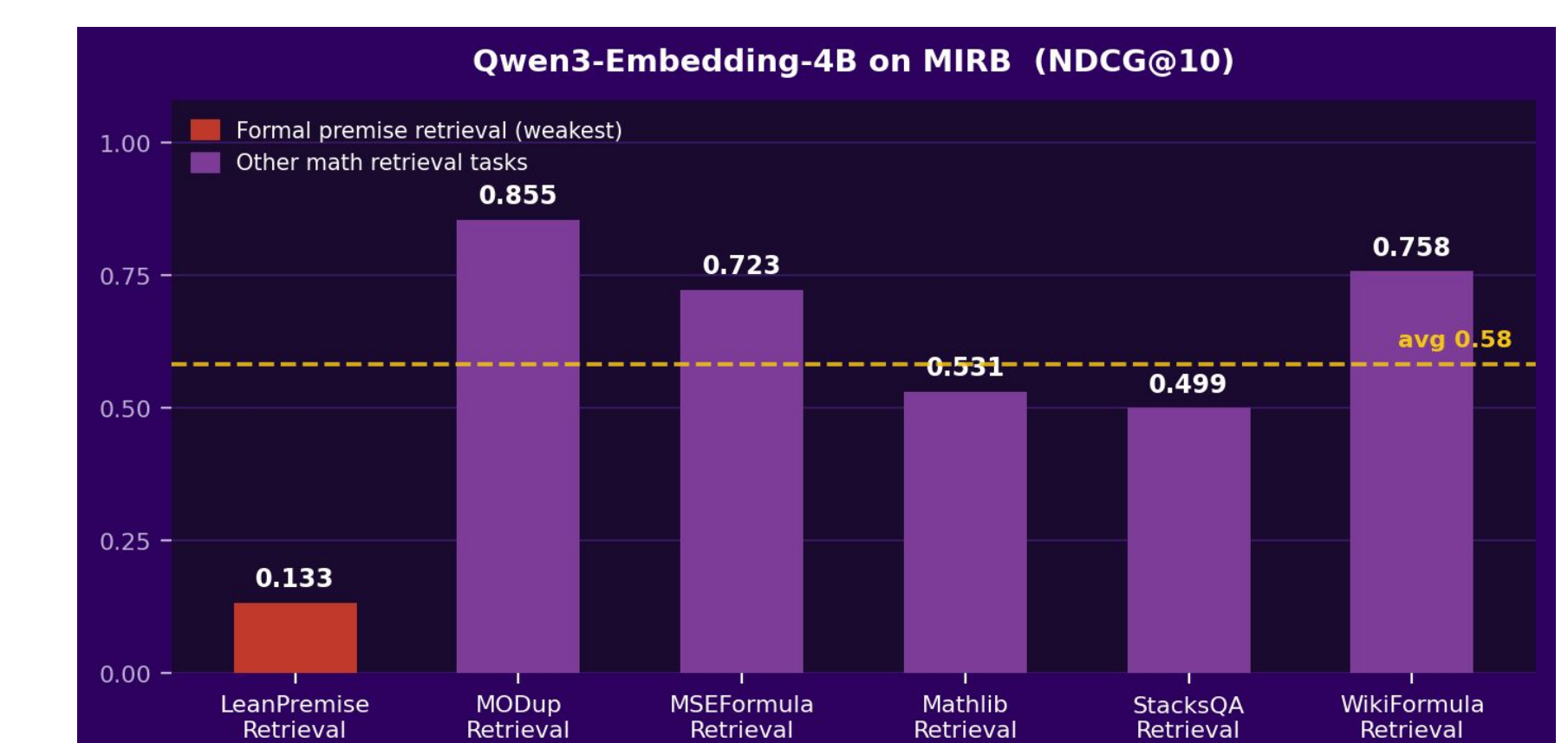
- **Blueprints:** Lean/natural language pairs with entailment
- **Mathematically equivalent theorems (new contribution!):** A growing dataset that uses basic mathematical equivalences to create translations of theorems in natural language



How Is the Performance of Existing Models?

General-purpose embedding models can achieve decent performance on natural language-based tasks

Performance of Qwen3-embedding-4b on the MIRB benchmark. Note that performance is worst for the Lean premise retrieval task.



...But off-the-shelf embedding models also have limitations

- The correspondence between embeddings of natural text and Lean is weak
- They do not appear to capture mathematical equivalence beyond subject similarity and keywords

Model/Task	Natural text to Lean, Recall@1	Lean to natural text, Recall@1	Vector space to module theory, recall@5	Probability theory to measure theory, recall@5
Qwen3-embedding-0.6B	0.375	0.369	0.286	0.200
Qwen3-embedding-4B	0.543	0.454	0.171	0.229
Qwen3-embedding-8B	0.573	0.518	0.314	0.314
Llama-embed-nemotron-8b	0.6562	0.5156	0.3429	0.3429

First two columns measure correspondence between natural text statements and Lean, the second two are small studies on the ability of embedding models to recognize mathematically equivalent statements that are phrased in the language of different subfields.

Next Steps

- Continued analysis of existing model performance. *Where exactly do they fail?*
- Use of these observations to better design of training routine for math specific embedding models